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Conceptual design and realization of a graphical tool for determining the test coverage of digital circuit diagrams

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vorgelegt von

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Abstract

According to Printed circuit board (PCB)s and their digital circuits which are getting more complex all the time there is the requirement of being able to determine the test coverage of it. In dependency on these determined values then automatic boundary scan chains could be performed or the need of changing something on the layout will be visible. This document describes the way of the conceptual design and the realization of a graphical tool for loading, showing different digital circuit diagrams projects which need to be in the Open database (ODB)++ format and the determination of test coverage parameters as well as the creation of Serial vector format (SVF) files for being able to run the boundary scan chain on these special devices which contains at least one Joint test action group (JTAG) device. Furthermore there will be described detailed information about the frameworks, programming patterns and best practices which were used for creating this Model-view-viewModel (MVVM) C# project and the project structure itself which was created for making sure to have the following key features available:

- Opening any ODB++ project by connecting to a stand alone Google remote procedure calls (GRPC) server and using the PCB-Investigator API
- Exporting any opened projects to Java script object notation (JSON) file for later usage and importing of that created files
- Creation and loading of project files which contains a manifest file for information about the project structure itself
- Modifying, exporting and importing of different kind of settings according the connection to the GRPC server, the appearance and the test coverage values
- Determination of different test coverage values according to the boundary scan mechanism
- Creation of SVF files for run the boundary scan chain with a programmer from ProMik GmbH on a JTAG containing device

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Chapter 1.

List of abbreviations

PCB	Printed circuit board
ODB	Open database
JTAG	Joint test action group
SVF	Serial vector format
GRPC	Google remote procedure calls
JSON	Java script object notation
MVVM	Model-view-viewModel
MCU	Micro controller
GUI	Graphical user interface
CPU	Central processing unit
RAM	Random access memory
GB	Gigabyte
CAD	Computer-aided design
API	Application interface
BSDL	Boundary scan description language
WPF	Windows presentation foundation
XAML	Extensible application markup language

Chapter 2.

Introduction

Within this chapter the motivation will be described, the requirements, the used programming language and operating system for creation of the whole project. All these aspects were previously discussed with the company to make sure to create a most valuable software for them.

2.1. Motivation

As the company ProMik GmbH is already providing programmers for being able to write data into different Micro controller (**MCU**)s now there came up the requirement, in dependency on customer demands, to perform boundary scan chains on **JTAG** devices. For being able to perform these kind of tests there should be a software available which allows to determine first the test coverage parameters in dependency on the boundary scan mechanism and create such **SVF** files for playing them with the programmer. This software should be provided as a complex framework where the opportunity is given to extend it in the future in elementary way by using much interfaces within the **MVVM** pattern.

2.2. Requirements

In order to provide all features as needed there were a couple of requirements which popped up at the very beginning of the conceptual design and during the implementation of the software:

- The software itself should consist of a Graphical user interface (**GUI**) application and a separate **GRPC** server for using the **PCB**-Investigator API
- There should be the possibility of opening **ODB++** project folders and display the result of circuit diagram in a useful way

- There should be the possibility of moving the visible area by mouse movement and zooming in and out of the visible area
- The visibility of each component should be settable
- All connections of one component to the others should be displayable
- The whole view of all components could be mirrorable along both axes
- The net names should be settable to visible and hidden
- An export mechanism of an opened project into **JSON** file for later usage should be available as an according import mechanism to be independent from the **PCB-Investigator** API as this requires a separate license for usage
- Settings should be available, exportable and importable
 - General settings according the appearance of the loaded project view
 - **GRPC**-Server related connection settings
 - General settings according the request of **PCB-Investigator** API
 - Test coverage related settings for determining all components for the boundary scan chain
 - **SVF** related settings according the programmer connection parameters to use
- Determination of test coverage parameters using the boundary scan mechanism
- Export of a file containing all pin information of all **JTAG** devices
- Creation of **SVF** files containing all instructions to be performed by the programmer while playing the boundary scan chain on a **JTAG** compatible device
- In order to create an extended-friendly framework there was furthermore the need of usage of company internal libraries instead of creating complete new ones
 - The usage of log panel was required
 - The usage of setting panel was required
 - The usage of **SVF** player was required

2.3. Programming language and operating system

The whole project was implemented with the software development environment called **Microsoft Visual Studio 2019** and the programming language **C#**. This is a quit powerful and modern language for creation of different desktop applications. The developer environment is really comfortable too, as it provides a NuGet package manager for organizing all external included libraries. Furthermore the whole project was created on a operating system **Windows 10 Professional x64** using a Central processing unit (**CPU**) of type **Intel(R)Xeon(R) E3-1240 v6 @ 3.70 GHz** and a total amount of Random access memory (**RAM**) of **16 Gigabyte (GB)**. This system makes sure to be able to implement without any delays of waiting for background processes at all and ensures to be able to run the developed software almost immediately without any delay times.

Chapter 3.

Fundamentals

Within this chapter the fundamentals regarding the later usage for implementation of the project is being described.

3.1. ODB++ project format

The **ODB++** format is a de facto standard format for creation of Computer-aided design (**CAD**) related projects[1] within the circuit board industry. This format itself was released in 1995 and after the component names were added then the suffix ++ was added in 1997. It hierarchically contains all necessary information about the digital circuit diagram in detail:

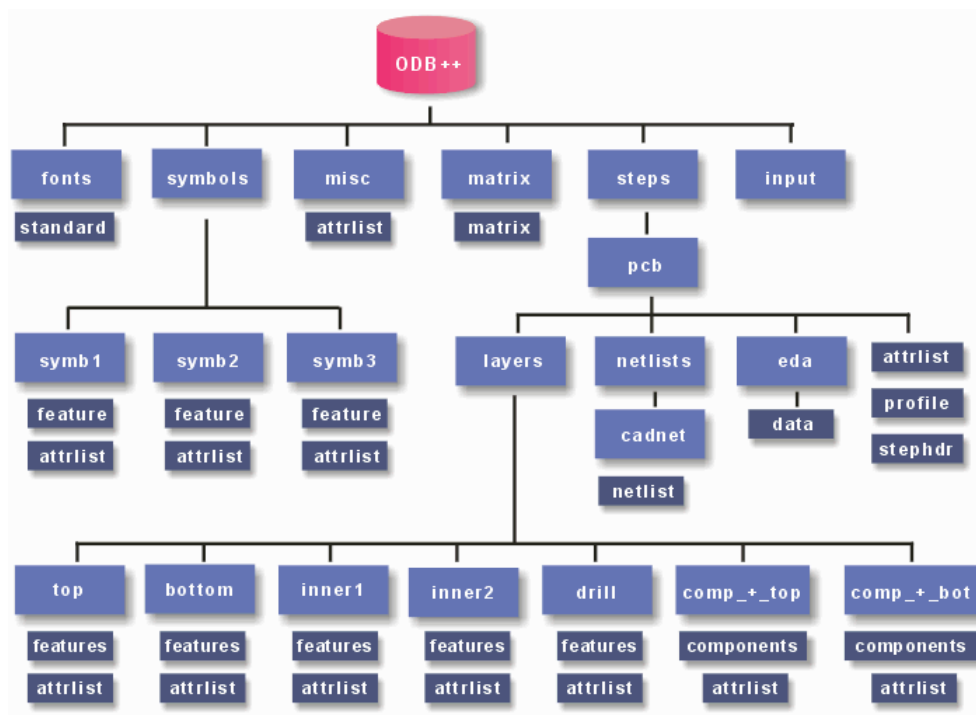


Figure 3.1.: ODB++ project structure[2]

As shown in 3.1 it is clear that this format contains all necessary components, net lists and attribute lists which are necessary for later determination of the test coverage of the circuit diagrams. These data will give detailed information regarding the placed components, their names, positions, sizes and connection to all nets. In dependency on the net connection the connected components could then be determined

3.2. PCB-Investigator

The **PCB-Investigator** is a software provided by EasyLogix also known as Schindler & Schill GmbH and provides many functionality regarding the opening, viewing and measuring of different kind of **CAD** related project files of digital circuit diagrams[3]:

- **ODB++**
- Gerber 274x
- Gen Cad 1.4
- Eagle files
- IDF files
- DXF files
- DPF files
- IPC2581 files
- IPC356 files
- PDF

It is a quit powerful tool which further provides the possibility to export almost all project types into another one and generate different kind of lists containing component relevant information as the component name and its pin information. Further the executable file application itself provides different public functions which are accessible by adding the **PCB-Investigator.exe** as a reference to the project.

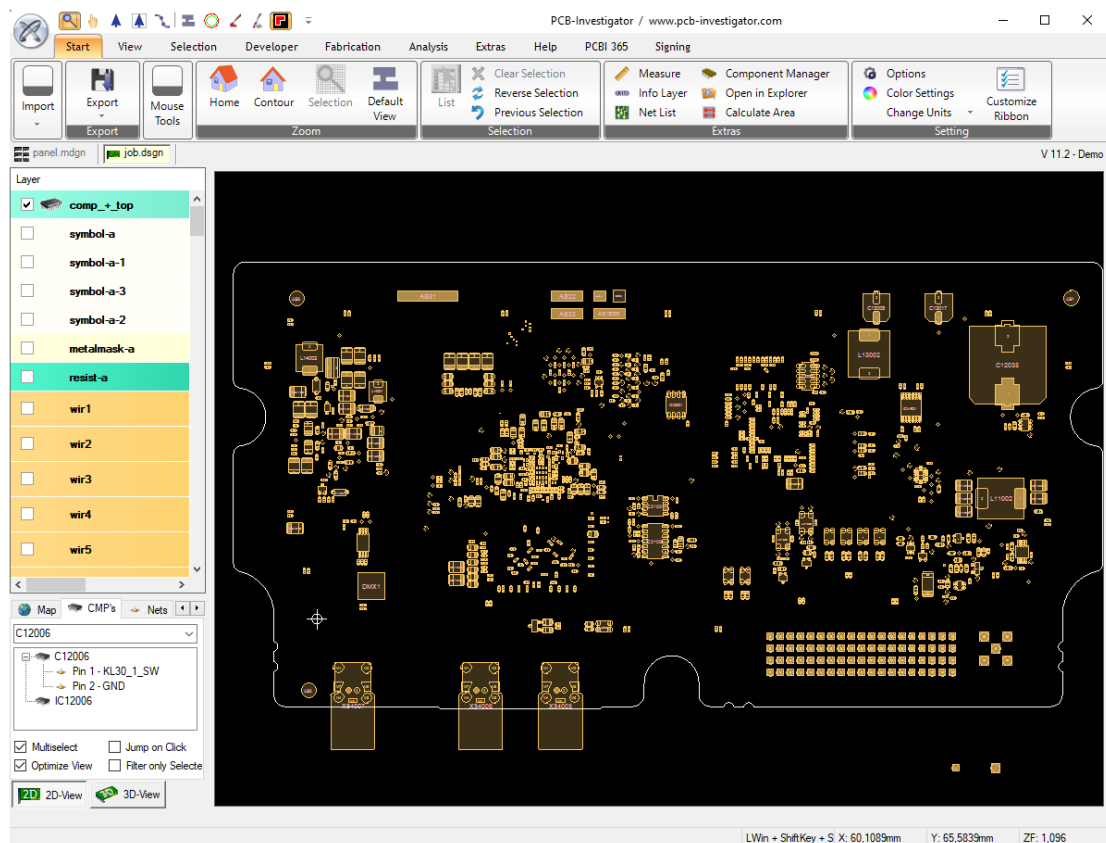


Figure 3.2.: PCB-Investigator overview

As pictured in 3.2 it can be seen that there are many possibilities of taking action with a loaded project. For verification purposes the most important feature was the CMPs tab where it is possible to search for any specific component in dependency on its name and to have a closer look in its consisting pins.

3.3. JTAG interface

The **JTAG** is a serial interface which is widely used for different field programmable gate arrays, complex programmable logic device configurations, the programming and debugging of different micro controllers. It consists of the following components in general:

- Test access port
 - Pin TDI - data in

- Pin TDO - data out
 - Pin TMS - control
 - Pin TCK - clock
 - Pin TRST - test reset
- Instruction register
 - Data register
 - TAP controller

The TAP controller is a state machine which could take in six valid different states[4]. This standard established within 1985/1986 and is being normed by IEEE 1149.11990. Later on by the overworked norm IEEE 1149.11994 the Boundary scan description language (**BSDL**) was considered as a default feature. The most actual version of the norm is IEEE 1149.1-2001. If there are more TAPs connected together then it will be called a scan chain in general.

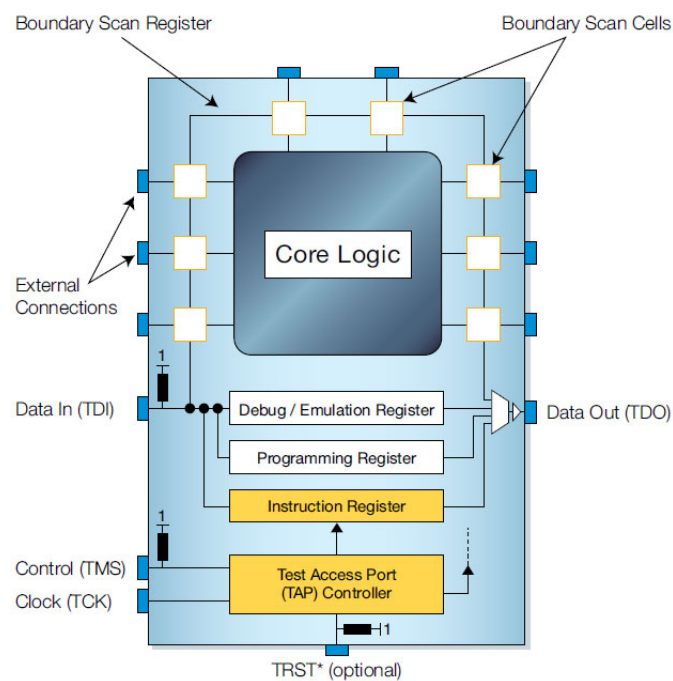


Figure 3.3.: JTAG interface in detail[5]

As shown in 3.3 it can be seen that that the **JTAG** device itself already contains some addition core logic for being able to perform different kind of test-debugging functions.

The boundary scan register itself contains a various amount of boundary scan cells which could be used for the testing logic.

3.4. Boundary scan

The main idea of the boundary scan is the purpose of ensuring that every component is mounted correctly in physically way as described within the circuit diagram[6]. In detail that means that according to each external pin connection there is in general the possibility to set any value to high or low and read back the physically present value of wiring. So if it is known for example that on a specific pin position an according pull down resistor is being connected then the further knowledge is that in every case the read back value should be low. Additionally there can be assumed that if a pull up resistor is being connected to a pin independently on the set value the read back value should be high. Finally the fact can be accepted that if some other components are connected to any pin which are not belonging to the pull down nor the pull up resistors then the read back value should be that one which was previously set. To perform a complete test of a device the following steps must be followed:

1. The clock rate must be triggered as long as the complete boundary scan register length totally is[7]
2. During that time the TDI data will be forwarded into each boundary scan cell
3. If all cells are covered by a value then the TMS pin should be triggered for passing all read values into the connected pins
4. The clock rate must be triggered again for the total length of the boundary scan register
5. During that time all values from the external pins are being read out and transmitted into the TDO
6. Now the read out values could be compared with some expected values to detect differences

3.5. Boundary scan description language

The boundary scan description language is a language for describing the boundary scan test ability of different **JTAG** compatible devices. It is based on the very high speed integrated circuit hardware description language and normed with IEEE 1149.1-2001. Typically these information are being provided by the manufacturer of the devices in a

.bsdl file for each component[8]. The files themselves are usually containing the following sections:

- port - the information about all pins and their usage mode (in, out, inout, linkage...)
- attribute PIN_MAP - the mapping from the used names within the port section to the real pin names used within the ODB++ project (position is important for later usage)
- attribute INSTRUCTION_LENGTH - instruction register length
- attribute BOUNDARY_REGISTER - all boundary scan compatible pins and their functionality (control, bidirectional, observe_only, output2)
- attribute BOUNDARY_LENGTH - the amount of boundary scan cells

3.6. SVF file format

The serial vector format is a format for communication with boundary scan compatible test vectors. It was introduced by Texas Instruments in 1991. The main purpose of the format is to ensure that independently on the used JTAG device different test operations could be performed successfully[9]. The format contains a couple of different instructions which could be performed by the boundary scan logic. The most important one according to this project is the **SDR length [TDI (tdi)] [TDO (tdo)][MASK (mask)] [SMASK (smask)]**; SDR means scan data register and tells the test to test all data register cells. The TDI is the data which should be passed into the registers[10]. TDO is the expected data to be read back from the data registers. And the MASK is the information about the positions of interest which should be considered. A valid instruction example is

SDR 24 TDI(000010) TDO(818181) MASK(FFFFFF);

In that specific example we have a total length of boundary scan cells of 24 and therefore the amount of passed hexadecimal coded values is 3 ($24 / 8$) -> 00 00 10 and 81 81 81 and FF FF FF. Further the TDI value 000010_{16} means that only on position 16 the value should be set to high ($10_{16} \rightarrow 16_{10}$). The TDO value 818181_{16} gives the information about the expected positions to be high after read back the data ($818181_{16} \rightarrow 100000011000000110000001_2$). That signifies only the positions 1, 8, 9, 16, 17 and 24 (read from right to left) should be set to high as the needed result. Finally the MASK value $FFFFFF_{16}$ shows that all positions of the 24 bit register should be considered.

3.7. Model-view-viewModel

The Model-view-viewModel pattern came up with the introduction of the Windows presentation foundation (**WPF**) framework from Microsoft and is designed for loose coupling programming with different components which are working together at the final stage. It could be seen as an enhancement of the model view controller pattern from JAVA programming language [11].

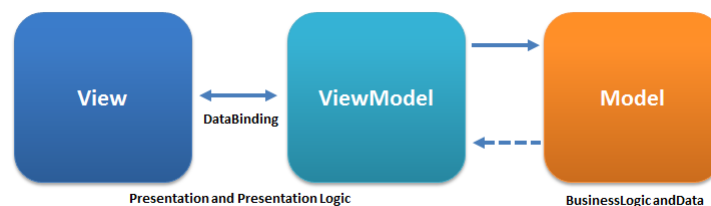


Figure 3.4.: MVVM structure[12]

As shown in 3.4 there are the three main components involved:

- View - the graphical representation form typically created in Extensible application markup language (**XAML**) code
- Model - content of all data to be used and business logic for it
- ViewModel - the connection layer to the view and contains representation logic

Typically there will be defined a various amount of data bindings in the view to the viewmodel. Further the view model itself informs the view about happened changes of values to be displayed by using the `INotifyPropertyChanged` event. Onward it is responsible for the connection to the data of any model and controls the function sequence to be performed in dependency on the user inputs which are being forwarded as `ICommand` objects. Finally the view should contain as less behind code as possible to make sure a high loose coupling is being ensured. The looser the coupling the easier it is to switch between new implementations of different components and replace them by others. In that way it should be quit easy possible to work on all three component types separately, without having to know of each others and finally connect them together.

3.8. GRPC

GRPC is a modern open source way for performing remote procedure calls independently on the programming language used for a specific program and works as a client-server

structure[13]. That means in detail to use this protocol there is the need first of defining a specific service file using the interface definition language. By using this language there is no need to consider programming language specific dependencies. Afterward by using protoc compiler there is the possibility of automated creation of the source code for a specific programming language which could then be included to finalize the communication work. Finally on one side there should be defined the logic for being a server by additionally defining a specific port to be used and on the other there needs to be defined the logic for being the client with considering the IP address of the server and the port. Using **GRPC** ensures bidirectional streaming possibilities and passes all object related data as a auto serialized stream in the **JSON** format. Further it can communicate through complex networks without having the challenge of defining more complex code for it. To define the service file there are many of elementary data types available, the possibility of defining object structures, functions and parameters or return values. A valid example of such a definition could be:

```

1 syntax = "proto3";
2
3 service PcbInvestigatorService {
4     // Gets all objects
5     rpc GetConvertedObjects(Request) returns (Result) {};
6 }
7
8 message Request {
9     int32 requestNumber = 1;
10    ComponentTypeGrpc type = 2;
11 }
12
13 message Result {
14     bool status = 1;
15 }
16
17 enum ComponentTypeGrpc {
18     UnknownComponentType = 0;
19     Arc = 1;
20 }

```

Listing 3.1: Interface definition of GRPC exchange

As shown in 3.1 there are all necessary values defined in structured and easy understandable way. Typically the auto generated code contains new classes which must be inherited for adding customized logic.

3.9. JSON format

The java script object notation is a serialized form of data output in textual form[14]. It is widely used where it could be necessary to understand and maybe modify data manually in an easy way. To work with that specific data format each programming language provides different kind of external libraries for parsing the information from objects into **JSON** and back from **JSON** to object. A valid **JSON** formatted text could like like:

```
1 {
2   "Pins": [
3     {
4       "Nets": [
5         0
6       ],
7       "Geometrics": {
8         "Bounds": {
9           "X": -1459,
10          "Y": 286,
11          "Width": 35,
12          "Height": 35
13        },
14        "CenterPoint": {
15          "X": 0.0,
16          "Y": 0.0
17        },
18        "Rotation": 0.0
19      },
20      "PinType": 2,
21      "PinNumber": "1"
22    }
23  ]
24 }
```

Listing 3.2: JSON formatted content

As shown in 3.2 the format is highly structured and build in hierarchical way.

Chapter 4.

Conceptual design

First of all the structure itself of the received objects needs to be analyzed more in detail which are being provided by the **PCB-Investigator.exe**:

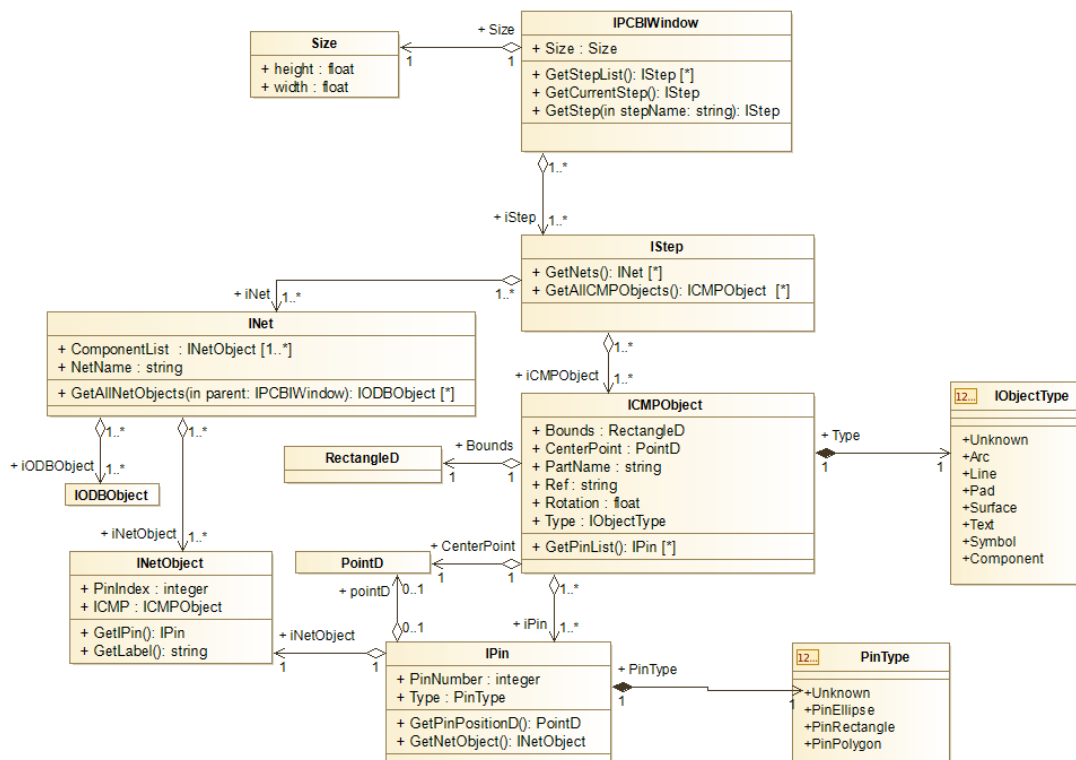


Figure 4.1.: Class diagram of the PCB-Investigator objects of interest

As shown within 4.1 there could be seen that the very first class is the IPCBIWindow which contains a various amount of IStep objects. Within this IStep objects there are many INet objects and ICMPObjets.

As ProMik GmbH is just interested in extracting all relevant information from any ODB++ related projects the following public functions of the **PCB-Investigator.exe** where used by the created software[15] to extract all INet and ICMPObjets:

- `IAutomation.IAutomationInit()` - Initialization of the Application interface (**API**)
- `IAutomation.CreateNewPCBIWindow(bool visible)` - Creates the base window
- `IPCBIWindow.LoadData(string path)` - Loads a specific project into the window
- `IPCBIWindow.GetStepList()` - Getter for all steps of the window
- `IStep.GetNets()` - Getter for all nets of a step
- `IStep.GetAllLayerNames()` - Getter for all layer names of a step
- `IStep.GetLayer(string layerName)` - Getter for a layer in dependency on the layer name from a step
- `ILayer.GetAllLayerObjects()` - Getter for all objects which are located on a specific layer within a step

Chapter 5.

Architecture

5.1. Overview

5.2. Interfaces

5.3. Grpc Server for using the PCB Investigator API

5.4. Grpc Client for connecting to the Server and parsing all objects

5.5. GUI

5.5.1. Events

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5.5.4. Implementations

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Chapter 6.

Analysis

- 6.1. Performance of loading ODB++ projects by using the PCBInvestigator API
- 6.2. Performance of loading projects by importing JSON formatted files
- 6.3. Performance of exporting projects into JSON format
- 6.4. Performance of showing connections of components
- 6.5. Performance of determination of test coverage values

Chapter 7.

Conclusion

Chapter 8.

Thanksgiving

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Appendix A.

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